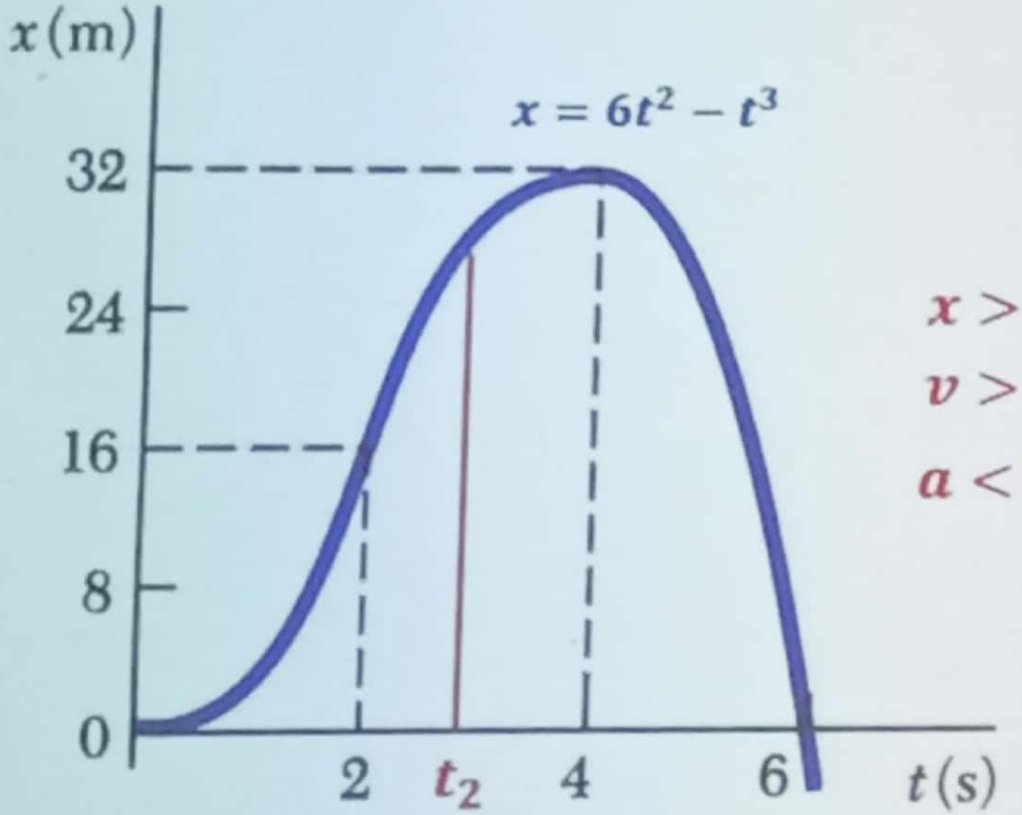
Rectilinear Motion: Analyzing the Position Graph

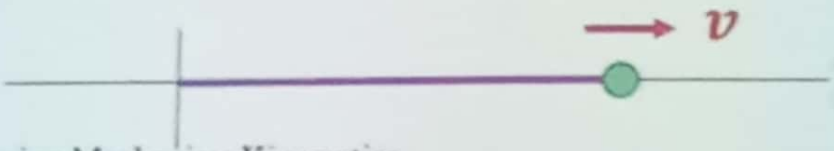


At time t_2:

$x > 0$, At the right

$v > 0$, Moving to the right

$a < 0$, Decelerating



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Rectilinear Motion

Position function:

$$x = 6t^2 - t^3$$

Velocity:

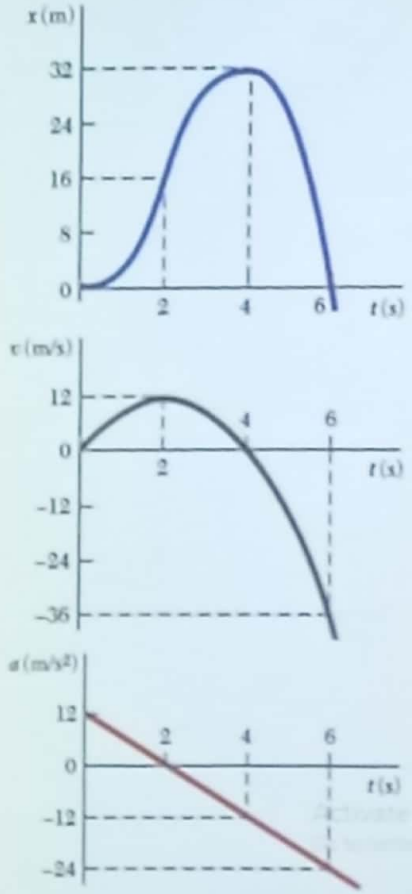
$$v = \frac{dx}{dt}$$

$$= 12t - 3t^2$$

Acceleration:

$$a = \frac{dv}{dt}$$

$$= 12 - 6t$$



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Kinematics

1. **Rectilinear Motion**
   a. Basic Relations
   b. **Constant acceleration**
   c. Variable acceleration

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Acceleration may be known in different forms:

a) Accel. is constant مسائل المرحلة الثانوية $a = a_c$

b) Accel. is a given function of time : $a = f(t)$

c) Accel. a given function of velocity : $a = f(v)$

d) Accel. a given function of displacement: $a = f(s)$

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Constant Acceleration ($a = a_c$)

$$a = \frac{dv}{dt} \longrightarrow dv = a_C dt$$

$$\int_{v_0}^{v} dv = a_C \int_{0}^{t} dt$$

At time $t_1 = 0$:
$v = v_0$,
At time $t_2 = t$:
$v = v$.

$$v]_{v_0}^{v} = a_C t]_0^t$$

$$v - v_0 = a_C t$$

$$v = v_0 + a_C t \qquad (1)$$

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a is a constant

$$v = v_0 + a_C t$$

$$v = \frac{ds}{dt} \longrightarrow ds = vdt$$

At time $t_1 = 0$: $s = s_0$,

At time $t_2 = t$: $s = s$.

$$\int_{s_0}^{s} ds = \int_{0}^{t} vdt = \int_{0}^{t} (v_0 + a_C t)dt$$

$$s]_{s_0}^{s} = (v_0 t + \frac{1}{2} a_C t^2)\Big|_0^t$$

$$s - s_0 = v_0 t + \frac{1}{2} a_C t^2 \quad (2)$$

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a is a constant

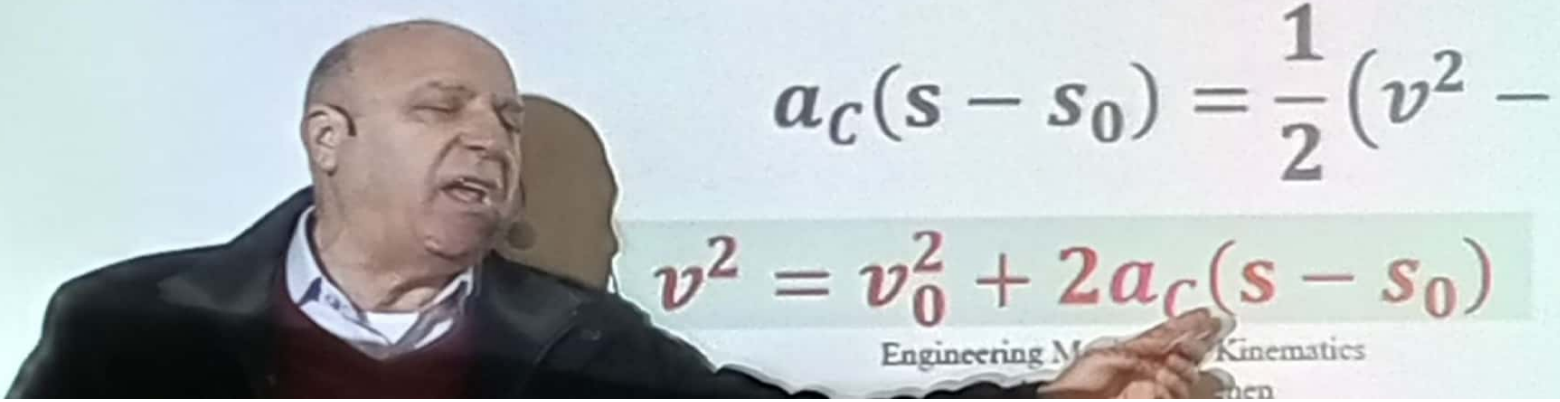
$$a_C = v\frac{dv}{ds} \longrightarrow ads = vdv$$

$$a_C \int_{s_0}^{s} ds = \int_{v_0}^{v} vdv$$

$$a_C s]_{s_0}^{s} = \left(\frac{1}{2}v^2\right)\Big|_{v_0}^{v}$$

$$a_C(s - s_0) = \frac{1}{2}\left(v^2 - v_0^2\right)$$

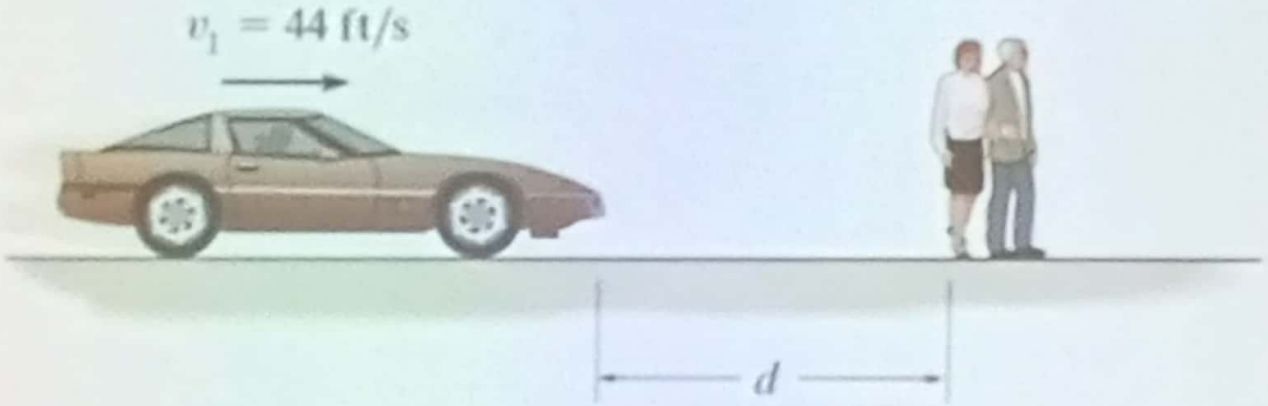
$$v^2 = v_0^2 + 2a_C(s - s_0) \qquad (3)$$



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Example:

A driver takes about 0.75 s before he or she can react to a situation to avoid a collision. If such drivers are traveling on a straight road at 30 mph (44 ft/s) and their cars can decelerate at 2 ft/s^2, determine the shortest stopping distance d from the moment the driver sees the pedestrians.



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Solution

(1) The car moves a distance S_0 before the driver decelerates the car, where

$$S_0 = vt$$

$$= 44(0.75) = 33.0 \text{ ft}$$

(2) The stopping distance:

$$(+\rightarrow) \qquad (v)^2 = (v_0)^2 + 2a(S - S_0)$$

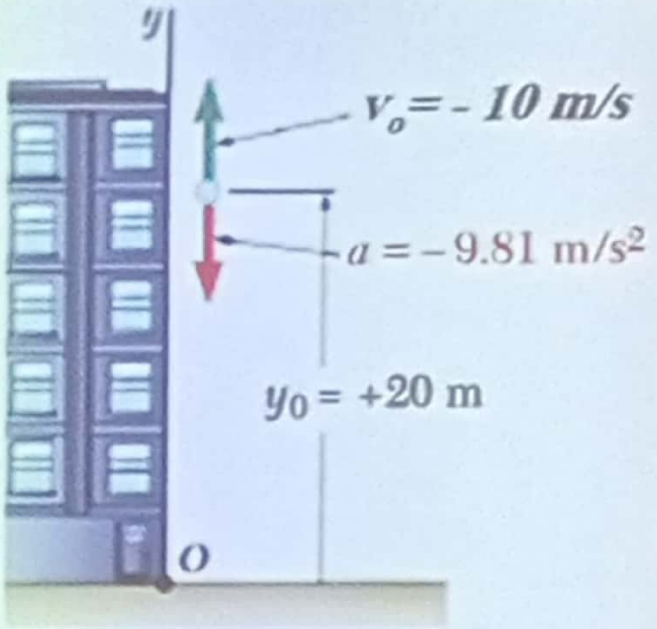
$$(0)^2 = (44)^2 + 2(-2)(d - 33)$$

$$\therefore \; d = 517 \text{ ft} \qquad \textit{Ans.}$$

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Example (Free fall)

A ball tossed with 10 m/s **upward** vertical velocity from window 20 m above ground.



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